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Palaeomagnetism of Dhar traps and drift of the subcontinent during the Deccan volcanism

G. V. S. Poornachandra Rao and M. S. Bhalla

Palaeomagnetism Group, National Geophysical Research Institute, Hyderabad – 500 007, India

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Summary. Palaeomagnetic investigations were carried out on nine lava flows around the Dhar region, which constitute the northern part of the Deccan traps. The stability of remanent magnetism of these specimens was analysed by an alternating magnetic field, thermal demagnetization and memory tests. Six flows exhibited characteristic components of magnetization, with a mean direction of $D = 143^\circ$, $I = +46^\circ$ ($K = 107.1$, $\alpha_{95} = 5.5^\circ$). This gives a VGP located at 29° N, 67° W ($\delta_p = 4.5^\circ$, $\delta_m = 7.0^\circ$). The lower site with normal magnetization and the upper five sites with reverse magnetization indicate a geomagnetic field reversal during the initial phases of Deccan volcanism in the Early Tertiary period. A rapid northward migration of about 18° in latitude and a simultaneous anticlockwise rotation of 37° is calculated for the subcontinent.

Introduction

Palaeomagnetic investigations on the Deccan traps were carried out by Clegg, Deutsch & Griffiths (1956), Deutsch, Radhakrishnamurthy & Sahasrabudhe (1959), Sahasrabudhe (1963), Bhimasankaram & Pal (1968), Athavale (1970), Athavale & Anjaneyulu (1971), Pal, Bindu Madhav & Bhimasankaram (1971), Wensink & Klootwijk (1971), Verma & Pullaiah (1971), Verma & Mital (1972, 1974), Kono, Kinoshita & Aoki (1972), Wensink (1973), Verma *et al.* (1973), Bhalla & Anjaneyulu (1974) and Bhalla & Poornachandra Rao 1974b) etc. However, very few of these studies were on traps from the northern part of the Deccan traps (Pal *et al.* 1971; Bhalla & Anjaneyulu 1974; Pal & Bhimasankaram 1971; Athavale 1970; Verma & Pullaiah 1971). In the present study results obtained from the Dhar area, constituting the northern part of the Deccan traps, are presented and implications of these results are discussed in terms of a prominent geomagnetic field reversal in the Early Tertiary period and rapid northward migration of the subcontinent during the period of Deccan volcanic activity (60–65 Myr ago, Wellman & McElhinny 1970 and Kaneoka & Haramura 1973).

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Geology, age and sampling

Deccan traps of the Dhar region constitute the northern part of main traps and have been considered to belong to the Middle Trap horizons (Pascoe 1964). The geology of the Dhar traps has been studied by the Geological Survey of India (Balasundaram 1970–71). The river Narmada flowing in an east–west direction divides the Peninsula of India and hence the Deccan traps into two tectonic regions (Choubey 1971). A number of individual flows and feeder dykes were identified and successive flows were separated by a 3–5 m thick redbole horizon suggesting that there were significant time intervals between their eruptions. From a study of polished sections it was inferred that magnetite and ilmenite are the main magnetic minerals and magnetite is altering to haematite along octahedral planes.

Several estimates of age of the Deccan traps have been made by stratigraphic (Krishnan 1960) and radiometric (Carnegie Institution of Washington 1963–64; Rama 1968; Wellman & McElhinny 1970; Kaneoka & Haramura 1973 etc.) methods. It was generally suggested that the Deccan traps erupted 60–65 Myr ago and the activity lasted for a relatively short period of time. On the basis of available radiometric age estimation of the Sagar traps (West 1969), the nearby Dhar traps are considered to be around 62 Myr old.

Nine separate flows were sampled between Dhar and Sendhwa at different altitudes (altitudes are given in Table 1) and the locations of the sampling sites are shown in Fig. 1. Fresh samples were collected from natural outcrops such as road cuttings, quarries and river

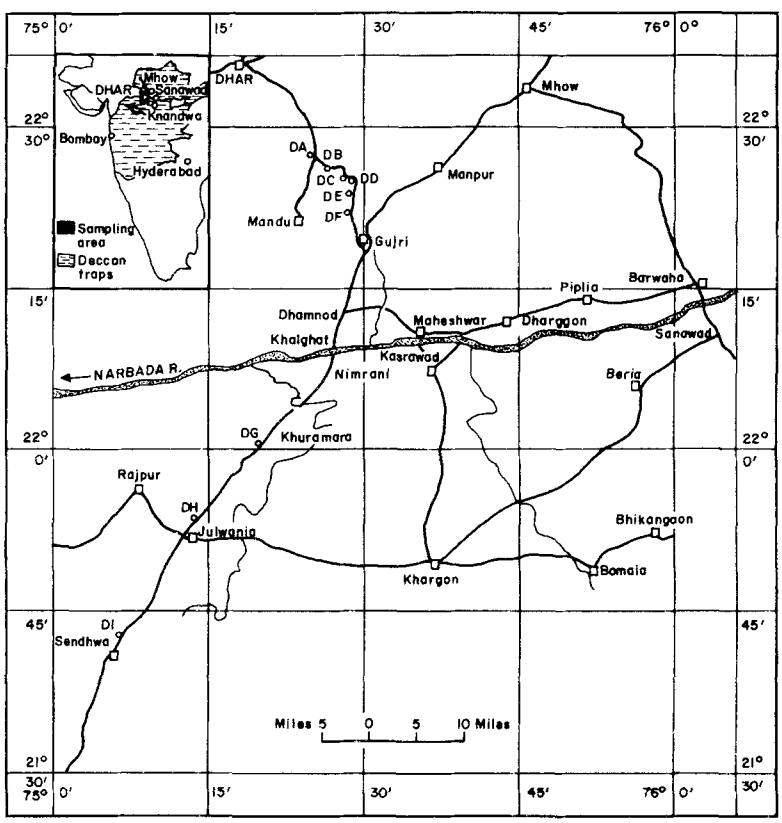


Figure 1. Location of sampling sites around the Dhar region, northern part of the Deccan traps.

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beds. The samples were oriented using a brunton compass and precision spirit level and the altitudes noted by means of a micro-altimeter which can be read to an accuracy of ±2 m. About five to six samples were collected from each site and 58 oriented blocks were sampled in total. In the laboratory the samples were cored and cut to obtain 230 cylindrical specimens of standard dimensions.

NRM measurement and stability tests

The remanent magnetic directions and intensities were measured on an astatic magnetometer and the stability has been tested by means of AF and thermal demagnetization and by magnetic memory techniques.

The NRM directions for sites DA and DB are reversed and for site DF are normal. Sites DD and DG show north–east declinations and down dips whereas the rest of the sites show scattered directions. The site mean intensities and susceptibilities range between 3.34–10.37×10^{-3} G and 1.59–5.72×10^{-3} G Oe^{-1} respectively. The mean Q_n ratios vary between 2.35 and 9.81. One specimen from each of the samples has been subjected to AF demagnetization in eight to ten steps up to a peak field of 400 Oe on a pilot basis. The partial AF treatment of the pilot specimens indicated that, mostly at fields of 100 and 200 Oe, the direction of the stable component is reached after removal of the secondary components and reveals the stable primary component of magnetization. Similar observations have been made on Deccan trap samples from several places by Athavale (1970), Athavale & Anjaneyulu (1971), Kono *et al.* (1972), Verma and Mital (1972), Verma *et al.* (1973) and Bhalla & Poornachandra Rao (1974b). Thus the partial AF treatment of specimens could to a large extent improve the grouping of remanent magnetic directions. Application of higher demagnetizing fields has not changed its direction any longer and it only showed a decrease in intensity. The stable directions obtained during these studies are not different from the directions obtained by cleaning in fields in excess of 500–1500 Oe (Pal *et al.* 1971; Wensink & Klootwijk 1971; Wensink 1973, Klootwijk 1974). Two of the sites (DH and DI) did not show any improvement either in grouping or stability. Therefore a few specimens from these sites were tested by partial thermal demagnetization. Even this experiment did not help in evaluating the characteristic component of magnetization and these two sites were therefore rejected. The characteristic components of magnetization for the Dhar region are shown in Fig. 2 obtained after demagnetization at a peak field of 100 Oe.

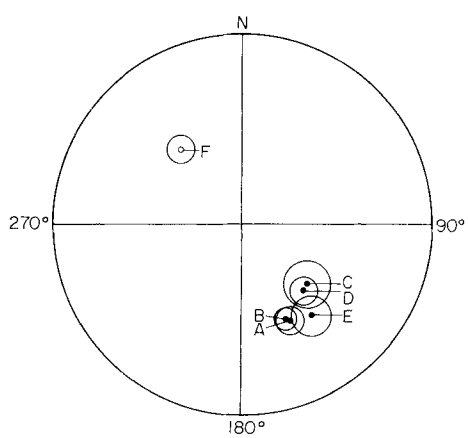


Figure 2. Site mean remanent magnetic directions of the Dhar traps along with their circles of confidence.

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Apart from these, some additional tests were conducted on the Dhar specimens (Bhalla & Poornachandra Rao 1974a) which indicated their suitability for palaeomagnetic considerations. These are low-field hysteresis and magnetic memory phenomenon tests (Radhakrishnamurthy 1970). These studies exhibited 'stable-line type Rayleigh loops', characteristic of the presence of single domain grains, indicating the acceptability of remanent magnetism carried by these rocks. About 53 per cent of the specimens could get through this critical test, revealing their stable component of magnetization. Most of the specimens from sites DH and DI have exhibited other types of Rayleigh loops due to the presence of interacting multi-domain grains, which have to be rejected straight away for palaeomagnetic consideration.

The coercive force (Hc) and thermomagnetic studies carried out on the Dhar specimens exhibit coercive forces ranging between 80–425 Oe and Curie points ranging between 544–623°C – characteristic of magnetite and suggesting that the probable carrier of remanance in these rocks is magnetite at different stages of oxidation, with varying amounts of titanium oxide.

Results

As a result of AF demagnetization at the selected peak field of 100 Oe, the specimens exhibit two directions of remanent magnetism. Site DF shows normal magnetization and sites DA, DB, DC, DD and DE show reverse magnetizations as shown in Fig. 2. Site DG shows a north–east declination and two sites, DH and DI, did not improve the grouping in AF and thermal studies. The site mean directions and their precision parameters are given in Table 1 before and after the AF treatment. When both the normal and reverse directions combined results in a mean direction of $D = 143°$, $I = +46°$, $K = 107.1$, $\alpha_{95} = 5.5°$ with a VGP located at $\lambda_p = 29°$ N, $L_p = 67°$ W ($\delta_p = 4.5°$, $\delta_m = 7.0°$). The mean direction and VGP from the Dhar traps agrees very well with those of lower Deccan traps from other localities. The implications of these results are discussed.

Early Tertiary geomagnetic reversal

The site mean remanent magnetism of the Dhar traps show two directions of magnetization, the lower normal (site DF) and the upper reverse (sites DA, DB, DC, DD and DE). These directions are compared with that of the lower Deccan traps in order to show an entirely different polarity inversion. The polarity sequences reported from several places of Deccan traps are shown in Fig. 3. As can be seen, in the lower Deccan traps the geomagnetic field changed its polarity from reverse to normal whereas in the Dhar traps it is from normal to reverse at an elevation of 390 m. It is in agreement with the reversal pattern at Malwa traps (Pal *et al.* 1971), which also constitute the northern part of the main traps. Similar observations of polarity inversion from normal to reverse has been made at Pohor and Ellichpur (Wensink 1973) and at Kundam (Verma *et al.* 1973).

Palaeomagnetic studies of Tirupati sandstones (Verma & Pullaiah 1967) which are considered to be Upper Cretaceous revealed a geomagnetic field reversal from normal to reverse. It was thought, apart from the reversal, that magnetic directions of the Tirupati sandstones are quite similar to the lower Deccan traps. The VGPs (Fig. 4) and the λ_ms (Table 2) of the Tirupati sandstones and the Dhar traps fall close to each other. Therefore, in the light of agreement of VGPs, λ_ms and polarities, the reversal in these two types of rocks can be correlated.

The pre-Deccan trap topography itself is found to be most rugged and the base of the Deccan traps is found to lie at different elevations at different places. However, from quite

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Table 1. Mean remanent magnetic directions and virtual geomagnetic pole positions of the Dhar traps.

Sl.	Site	Elevation		NRM				After AF demagnetization				Treatment	VGP position			
no.	no.	(m)	$N(n)$	D	I	K	α°_{95}	D	I	K	α°_{95}	(Oe)	$\lambda_p{}^{\circ}$N	$L_p{}^{\circ}$W	$\delta_p{}^{\circ}$	$\delta_m{}^{\circ}$
1	DA	584	5(27)	144	+54	8.3	21.6	153	+42	85.1	6.7	100	36	74	5.0	8.2
2	DB	560	6(19)	159	+49	69.2	6.8	154	+43	89.2	6.0	100	36	75	4.6	7.4
3	DC	490	5(15)	81	+67	14.3	16.5	132	+51	29.7	12.8	100	20	63	11.7	17.3
4	DD	502	6(34)	54	+30	11.8	16.6	137	+50	139.3	4.8	100	23	65	4.3	6.4
5	DE	448	5(14)	24	+64	22.0	13.3	141	+39	47.6	9.1	100	31	61	6.5	10.8
6	DF	381	10(48)	329	+42	47.5	6.4	320	−48	59.2	6.1	100	26	66	5.2	8.0
7	DG	225	5(27)	27	+46	65.3	7.7	35	+41	58.8	8.1	100	58	156° E	6.0	9.8
8	DH	280	6(24)	346	+57	4.9	25.6	Scattered								
9	DI	377	6(23)	263	+54	2.5	35.5	Scattered								
	Mean of DA – DF							143	+46	107.1	5.5		29	67	4.5	7.0

N = Number of samples.
n = Number of specimens.
D = Declination east of north.
I = Inclination of +ve down and −ve up.
K = Precision parameter.

α_{95} = Radius of circle of confidence at 95 per cent probability level.
λ_p = Latitude and longitude of VGP.
L_p =
δ_p = Semi-axes of the elliptical error around.
δ_m = the pole at 95 per cent probability level.

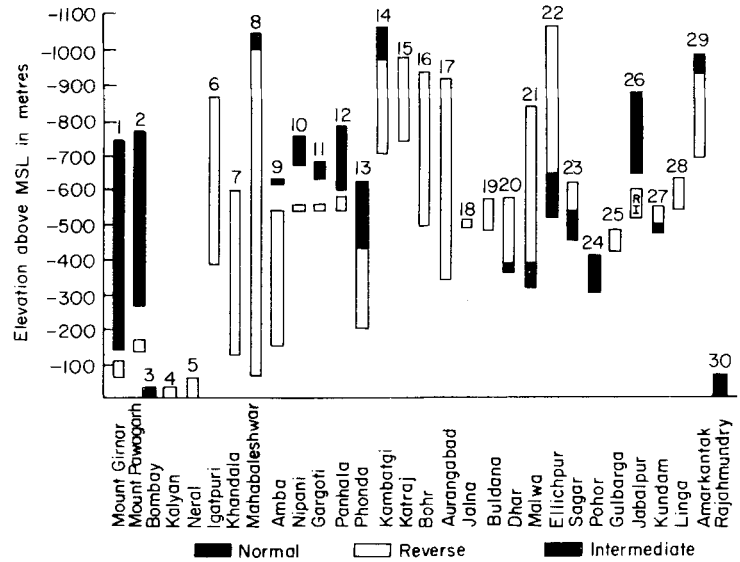


Figure 3. Correlation of geomagnetic polarity sequences from various localities of the Deccan traps along with the Dhar traps.

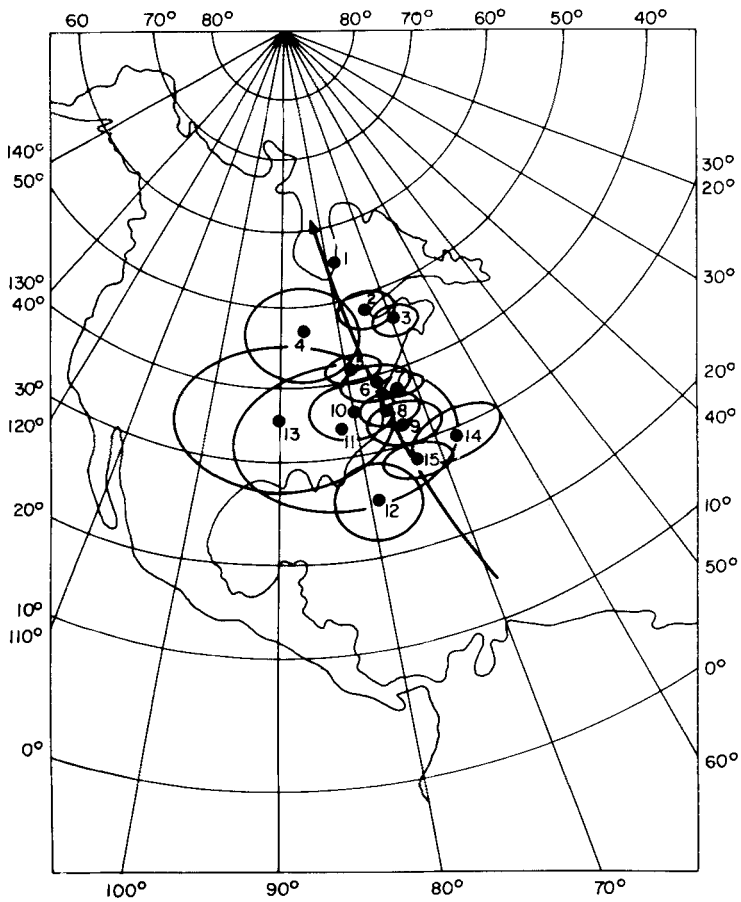


Figure 4. Apparent polar wandering curve for India during the Upper Cretaceous–Lower Tertiary period.

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Table 2. Palaeomagnetic parameters from some vertical sequences of the Deccan traps.

Sl. no.	Locality	Mean D	Mean I	Polarity	α_{95}°	λ_m°S	Pole position λ_p°N	Pole position L_p°W	Pole position δ_p°	Pole position δ_m°	References
1	Mahabaleshwar (uppermost)	344	−25	N	–	12	55	78	–	–	Sashasrabudhe (1963)
2	Jabalpur	343	−28	N	5.0	17	48	74	3.0	5.5	Verma & Pullaiah (1971)
3	Jabalpur-Dindori	340	−31	N	4.5	18	46	69	2.8	5.0	Verma *et al.* (1973)
4	Mahabaleshwar (upper)	345	−40	N	10.0	21	47	86	7.3	12.0	Sahasrabudhe (1963)
5	Mount Girnar	336	−38	N	4.4	25	41	79	3.0	5.2	Verma & Mital (1972)
6	Mount Pavagarh	334	−38	N	5.7	25	39	75	3.9	6.7	Verma & Mital (1974)
7	Sagar	156	+40	R	3.0	25	38	73	2.2	3.6	Bhalla & Anjaneyulu (1974)
8	Western Ghats	152	+50	R	3.8	30	34	76	3.4	5.1	Wensink & Klootwijk (1971)
9	Aurangabad	150	+48	R	5.5	30	33	73	4.7	7.2	Athavale & Anjaneyulu (1971)
10	Ellichpur	159	+48	R	5.3	30	36	80	4.5	6.9	Wensink (1973)
11	Mahabaleshwar (lower)	157	+52	R	11.0	32	34	82	10.3	15.0	Sahasrabudhe (1963)
12	Khandala	147	+58	R	5.0	39	25	79	5.4	7.4	Sahasrabudhe (1963)
13	Malwa	164	+49	R	11.9	32	36	90	10.3	15.6	Pal *et al.* (1971)
		353	−54	N							
14	Dhar	144	+45	R	5.5	30	29	67	4.5	7.0	Present study
		320	−48	N							
15	Tirupati Sandstones	153	+56	R	4.0	34	28	73	4.1	5.8	Verma & Pullaiah (1967)
		329	−59	N							

λ_m = palaeolatitude of Nagpur; other symbols are as in Table 1.

an extensive study of elevations over the Archaeans, the Bizawars and the Vindhyans, over which the Deccan traps in the Narmada valley are found to lie, a summit accordance is noticed at 330 m around the Dhar forest region by Choubey (1973). Hence the base of the Deccan traps around the Dhar region is considered to be at 330 m elevation. Therefore, on the basis of the above two considerations: (1) agreement of polarity sequence and VGPs of the Dhar and Tirupati sandstones, (2) considering the base of the traps at 330 m around Dhar region, the polarity transition observed at an elevation of 390 m at the Dhar and Malwa traps can be considered to be a prominent geomagnetic field reversal in the early phases of eruption of the Deccan traps.

Therefore this can be explained as follows. The Earth's field was initially normal at the commencement of the Deccan volcanic eruption, then changed to reverse during which time large volumes of rocks were poured out and then changed back to normal. This latter reversal is the generally observed one all over the traps at an elevation of 600 ± 60 m. The reversal recorded in the Dhar and Malwa traps and the Tirupati sandstones can be correlated with the earlier one (normal to reverse) during the initial phases of eruption in the Early Tertiary.

Drift of the subcontinent during Deccan volcanism

Several estimates of the amount and rate of drift of the Indian subcontinent have been made (e.g. Clegg *et al.* 1956; Clegg, Rhadakrishnamurthy & Sahasrabudhe 1958; McElhinny 1968; Wensink 1973, 1975; Verma *et al.* 1973; Peirce 1978; Duncan 1978; Klootwijk 1979 etc.) and data from the Deccan traps suggest a rapid northward migration. The igneous activity represented by the Dhar traps is correlated with those from other parts of the Deccan traps. The palaeomagnetic results of some of the Deccan traps and Tirupati sandstones are listed in Table 2 which covers the Upper Cretaceous–Lower Tertiary. The VGPs of these rocks are plotted on equal area projection which gives the apparent polar wandering curve for the subcontinent as shown in Fig. 4. The successive VGPs from all these sections reach the present north pole with the VGPs of the Dhar traps and Tirupati sandstones being south of all the others and that of the Mahabaleswar (uppermost) normal traps being nearer. From this it is clear during the eruption of different flows that the VGP shows a shift from 25° N to 55° N. As such a rapid wandering of the pole is not possible, this can be interpreted as a drift of the subcontinent amounting to 18° in latitude associated with an anticlockwise rotation of 37° during the entire Deccan volcanism.

In a recent review of palaeomagnetic data of the Indo-Pakistani fragment of Gondwanaland, Klootwijk (1979) argued that during the Deccan volcanism the subcontinent had not moved more than 5° in latitude. Although the Deccan traps constitute the most throughly studied formation in the world of palaeomagnetic literature, the two basic questions of many more reversals of the geomagnetic field and rapid northward movement of the landmass remained disputable.

Conclusions

The Dhar traps are uniform in their magnetic character with magnetite as the main carrier of remanence in them. They are the earlier phases of eruption of the Deccan traps with a lower normal and upper reverse magnetization representing an Early Tertiary geomagnetic field reversal. The VGPs obtained from several investigations on the Deccan traps migrate from 25° N to 55° N which is interpreted as a drift of the subcontinent of about 18° in latitude. In spite of thorough palaeomagnetic studies of the Deccan traps, basic questions like the rapid northward migration of India and several reversals of the magnetic field remained unsolved as a result of diverging views expressed by several workers.

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